Dynamic Node-Topology Virtual Network Embedding

March 20, 2026

1 Problem Formulation

1.1 Time-Varying Substrate Network

Unlike conventional VNE, which typically assumes a quasi-static substrate during the embedding process, we consider a spatio-temporal substrate network with both static and dynamic components. In particular, the set of candidate substrate-node locations is fixed over the planning horizon, whereas the set of active substrate nodes and links may vary with time.

Let $\bar{\mathcal{N}}^s$ denote the set of all candidate substrate-node locations, which is assumed to remain unchanged throughout the planning horizon. This candidate set is partitioned into two disjoint subsets:

$$\bar{\mathcal{N}}^s = \mathcal{N}^{\mathrm{st}} \cup \bar{\mathcal{N}}^{\mathrm{dy}}, \qquad \mathcal{N}^{\mathrm{st}} \cap \bar{\mathcal{N}}^{\mathrm{dy}} = \varnothing, \tag{1}$$

where $\mathcal{N}^{\mathrm{st}}$ is the set of static substrate nodes and $\bar{\mathcal{N}}^{\mathrm{dy}}$ is the set of candidate dynamic substrate nodes. The static nodes are permanently present in the substrate, whereas candidate dynamic nodes may become active or inactive over time.

At time t, the substrate network is represented by $\mathcal{G}^s(t) = \big(\mathcal{N}^s(t), \mathcal{L}^s(t)\big)$, where $\mathcal{N}^s(t)$ and $\mathcal{L}^s(t)$ denote the active substrate node set and the active substrate link set at time slot t, respectively.

For each candidate dynamic node $n \in \bar{\mathcal{N}}^{\mathrm{dy}}$, let $a_n(t) \in \{0, 1\}$ denote its physical availability at time slot t, where $a_n(t) = 1$ indicates that node n is physically available and $a_n(t) = 0$ otherwise. Accordingly, the active substrate node set at time slot t is given by

$$\mathcal{N}^s(t) = \mathcal{N}^{\mathrm{st}} \cup \left\{ n \in \bar{\mathcal{N}}^{\mathrm{dy}} \mid a_n(t) = 1 \right\}. \tag{2}$$

Each substrate node $n \in \bar{\mathcal{N}}^s$ is characterized by a finite amount of computational resources. Let $\overline{C}_n$ and $\overline{M}_n$ denote the maximum CPU capacity and the maximum memory capacity of substrate node n, respectively. These parameters specify the upper bounds on the amount of virtual-node resources that can be allocated to node n at any time slot.

The substrate topology also consists of a static part and a dynamic part. Let $\mathcal{L}^{\mathrm{st}}$ denote the set of static undirected links, and let

$$\bar{\mathcal{L}}^{\mathrm{dy}} \subseteq \left\{ (i, j) \mid i, j \in \bar{\mathcal{N}}^s,\ i < j \right\}$$

denote the set of candidate undirected dynamic links supported by optical switching or programmable interconnects. For each candidate dynamic link $\ell = (i, j) \in \bar{\mathcal{L}}^{\mathrm{dy}}$, let $q_\ell(t) \in \{0, 1\}$ denote its activation state at time slot t. Then, the active substrate link set is

$$\mathcal{L}^s(t) = \mathcal{L}^{\mathrm{st}} \cup \left\{ \ell \in \bar{\mathcal{L}}^{\mathrm{dy}} \mid q_\ell(t) = 1 \right\}. \tag{3}$$

A dynamic link can be activated only if both of its endpoints are operational. To unify static and dynamic nodes, define

$$\tilde{z}_n(t) = \begin{cases} 1, & n \in \mathcal{N}^{\mathrm{st}}, \\ z_n(t), & n \in \bar{\mathcal{N}}^{\mathrm{dy}}. \end{cases}$$

Then, for each $\ell = (i, j) \in \bar{\mathcal{L}}^{\mathrm{dy}}$,

$$q_\ell(t) \le \tilde{z}_i(t), \qquad q_\ell(t) \le \tilde{z}_j(t), \qquad \forall \ell = (i, j) \in \bar{\mathcal{L}}^{\mathrm{dy}},\ t \in \mathcal{T}. \tag{4}$$

For routing purposes, each undirected link is replaced by two opposite directed arcs. Let $\mathcal{A}^{\mathrm{st}}$ and $\bar{\mathcal{A}}^{\mathrm{dy}}$ denote the directed arc sets induced by $\mathcal{L}^{\mathrm{st}}$ and $\bar{\mathcal{L}}^{\mathrm{dy}}$, respectively, and define

$$\bar{\mathcal{A}}^s = \mathcal{A}^{\mathrm{st}} \cup \bar{\mathcal{A}}^{\mathrm{dy}}. \tag{5}$$

For each directed arc $(i,j) \in \bar{\mathcal{A}}^s$, let $\bar{B}_{ij}$ denote its bandwidth. Its activation state is defined as

$$y_{ij}(t) = \begin{cases} 1, & (i,j) \in \mathcal{A}^{\mathrm{st}}, \\ q_{(i,j)}(t), & (i,j) \in \bar{\mathcal{A}}^{\mathrm{dy}}. \end{cases} \tag{6}$$

1.2 Virtual Request Model

Let $\mathcal{R}$ denote the set of all virtual network requests within the planning horizon. Each request $r \in \mathcal{R}$ is characterized by an arrival time slot t_r^a and a departure time slot t_r^d, where $t_r^a \le t_r^d$. Throughout its service duration, request r is modeled as a virtual graph

$$\mathcal{G}_r^v = \left(\mathcal{N}_r^v, \mathcal{L}_r^v\right), \tag{7}$$

where $\mathcal{N}_r^v$ and $\mathcal{L}_r^v$ denote the sets of virtual nodes and virtual links, respectively.

Each virtual node $u \in \mathcal{N}_r^v$ is associated with a CPU demand $c_{r,u}$ and a memory demand $m_{r,u}$. Similarly, each virtual link $e = (u,v) \in \mathcal{L}_r^v$ is associated with a bandwidth demand $b_{r,e}$.

Accordingly, the set of active virtual network requests at time slot t is defined as

$$\mathcal{R}_t \triangleq \left\{ r \in \mathcal{R} \mid t_r^a \le t \le t_r^d \right\}. \tag{8}$$

1.3 Decision Variables

We introduce the following binary decision variables.

1) Request admission For each request $r \in \mathcal{R}$, define

$$\eta_r = \begin{cases} 1, & \text{if request } r \text{ is admitted}, \\ 0, & \text{otherwise}. \end{cases} \tag{9}$$

The admission variable η_r is determined when request r arrives and remains fixed throughout its lifetime. Hence, if $\eta_r = 1$, request r must be feasibly embedded for all $t \in [t_r^a, t_r^d]$; otherwise, the request is rejected and consumes no substrate resources.

2) Virtual-node mapping For each request $r \in \mathcal{R}$, each virtual node $u \in \mathcal{N}_r^v$, each candidate substrate node $n \in \bar{\mathcal{N}}^s$, and each time slot $t \in \mathcal{T}$, define

$$x_{r,u,n}(t) = \begin{cases} 1, & \text{if virtual node } u \text{ of request } r \text{ is mapped to substrate node } n \text{ at time slot } t, \\ 0, & \text{otherwise}. \end{cases} \tag{10}$$

3) Virtual-link mapping For each request $r \in \mathcal{R}$, each virtual link $e = (u,v) \in \mathcal{L}_r^v$, each directed substrate arc $(i,j) \in \bar{\mathcal{A}}^s$, and each time slot $t \in \mathcal{T}$, define

$$f_{r,e,ij}(t) = \begin{cases} 1, & \text{if arc } (i,j) \text{ is used in the routing path of virtual link } e \text{ of request } r \text{ at time slot } t, \\ 0, & \text{otherwise}. \end{cases} \tag{11}$$

4) Virtual-node migration To capture node migration over time, for each request $r \in \mathcal{R}$, each virtual node $u \in \mathcal{N}_r^v$, each pair of candidate substrate nodes $n, m \in \bar{\mathcal{N}}^s$ with $n \ne m$, and each time slot $t \in \mathcal{T}$, define

$$g_{r,u,nm}(t) = \begin{cases} 1, & u \text{ of request } r \text{ migrates from substrate node } n \text{ to substrate node } m \text{ at time slot } t, \\ 0, & \text{otherwise}. \end{cases} \tag{12}$$

5) Node and topology reconfiguration For each candidate dynamic substrate node $n \in \bar{\mathcal{N}}^{\mathrm{dy}}$ and each time slot $t \in \mathcal{T}$, let $\delta_n^+(t)$ and $\delta_n^-(t)$ denote the node power-on and power-off indicators, respectively. Likewise, for each candidate dynamic link $\ell \in \bar{\mathcal{L}}^{\mathrm{dy}}$ and each time slot $t \in \mathcal{T}$, let $\sigma_\ell^+(t)$ and $\sigma_\ell^-(t)$ denote the dynamic-link activation and deactivation indicators, respectively.

1.4 Cost, Energy, and Load-Balancing Model

1) Node utilization For each candidate substrate node $n \in \bar{\mathcal{N}}^s$, the CPU and memory utilization ratios at time slot t are defined as

$$u_n^{\text{cpu}}(t) = \frac{\sum_{r\in\mathcal{R}_t}\sum_{u\in\mathcal{N}_r^v} c_{r,u}\, x_{r,u,n}(t)}{\overline{C}_n}, \tag{13}$$

$$u_n^{\text{mem}}(t) = \frac{\sum_{r\in\mathcal{R}_t}\sum_{u\in\mathcal{N}_r^v} m_{r,u}\, x_{r,u,n}(t)}{\overline{M}_n}. \tag{14}$$

2) Dynamic node energy For each candidate dynamic substrate node $n \in \bar{\mathcal{N}}^{\text{dy}}$, the running energy consumption at time slot t is given by

$$E_n^{\text{run}}(t) = z_n(t)\Big(E_n^b + \kappa_n^c u_n^{\text{cpu}}(t) + \kappa_n^m u_n^{\text{mem}}(t)\Big), \tag{15}$$

where E_n^b denotes the base operating energy of node n, and κ_n^c and κ_n^m are the energy coefficients associated with CPU and memory utilization, respectively.

The node state-switching energy is defined as

$$E_n^{\text{sw}}(t) = \delta_n^+(t)E_n^{\text{on}} + \delta_n^-(t)E_n^{\text{off}}, \tag{16}$$

where E_n^{on} and E_n^{off} denote the energy consumed for powering on and powering off node n, respectively.

3) Dynamic topology energy For each candidate dynamic link $\ell \in \bar{\mathcal{L}}^{\text{dy}}$, let E_ℓ^{act} denote the operating energy of an active dynamic link, and let E_ℓ^{up} and E_ℓ^{down} denote the activation and deactivation energies, respectively. Then, the topology-related energy consumption at time slot t is

$$E^{\text{top}}(t) = \sum_{\ell\in\bar{\mathcal{L}}^{\text{dy}}} \Big(q_\ell(t)E_\ell^{\text{act}} + \sigma_\ell^+(t)E_\ell^{\text{up}} + \sigma_\ell^-(t)E_\ell^{\text{down}}\Big). \tag{17}$$

4) Migration cost Let $s_{r,u}$ denote the migration state size of virtual node u in request r, and let τ_{nm}^{mig} denote the migration penalty between substrate nodes n and m. Then, the migration cost at time slot t is defined as

$$C^{\text{mig}}(t) = \sum_{r\in\mathcal{R}_t}\sum_{u\in\mathcal{N}_r^v}\sum_{\substack{n,m\in\bar{\mathcal{N}}^s\\ n\neq m}} g_{r,u,nm}(t)\big(\xi_1 s_{r,u} + \xi_2\tau_{nm}^{\text{mig}}\big), \tag{18}$$

where $\xi_1, \xi_2 \geq 0$ are weighting coefficients.

5) Resource consumption cost In addition to energy-related costs, we consider the operational cost incurred by the actual amount of substrate resources consumed at time slot t. Let χ_c, χ_m, and χ_b denote the unit costs of CPU, memory, and bandwidth resource consumption, respectively. Then, the resource consumption cost at time slot t is defined as

$$C^{\text{res}}(t) = \chi_c \sum_{n\in\bar{\mathcal{N}}^s}\sum_{r\in\mathcal{R}_t}\sum_{u\in\mathcal{N}_r^v} c_{r,u}\, x_{r,u,n}(t) + \chi_m \sum_{n\in\bar{\mathcal{N}}^s}\sum_{r\in\mathcal{R}_t}\sum_{u\in\mathcal{N}_r^v} m_{r,u}\, x_{r,u,n}(t) + \chi_b \sum_{(i,j)\in\bar{\mathcal{A}}^s}\sum_{r\in\mathcal{R}_t}\sum_{e\in\mathcal{L}_r^v} b_{r,e}\, f_{r,e,ij}(t), \tag{19}$$

where χ_c, χ_m, and χ_b are the per-unit operational costs of CPU, memory, and bandwidth resource usage, respectively.

6) Link utilization and system load balance For each directed substrate arc $(i,j) \in \bar{\mathcal{A}}^s$, the bandwidth utilization ratio at time slot t is

$$u_{ij}^{\text{bw}}(t) = \frac{\sum_{r\in\mathcal{R}_t}\sum_{e\in\mathcal{L}_r^v} b_{r,e}\, f_{r,e,ij}(t)}{\overline{B}_{ij}}. \tag{20}$$

The average CPU, memory, and bandwidth utilization ratios are respectively defined as

$$\bar{u}^{\mathrm{cpu}}(t) = \frac{1}{|\bar{\mathcal{N}}^s|} \sum_{n \in \bar{\mathcal{N}}^s} u_n^{\mathrm{cpu}}(t), \tag{21}$$

$$\bar{u}^{\mathrm{mem}}(t) = \frac{1}{|\bar{\mathcal{N}}^s|} \sum_{n \in \bar{\mathcal{N}}^s} u_n^{\mathrm{mem}}(t), \tag{22}$$

$$\bar{u}^{\mathrm{bw}}(t) = \frac{1}{|\bar{\mathcal{A}}^s|} \sum_{(i,j) \in \bar{\mathcal{A}}^s} u_{ij}^{\mathrm{bw}}(t). \tag{23}$$

Accordingly, the CPU, memory, and bandwidth load variances are given by

$$L^{\mathrm{cpu}}(t) = \frac{1}{|\bar{\mathcal{N}}^s|} \sum_{n \in \bar{\mathcal{N}}^s} \left(u_n^{\mathrm{cpu}}(t) - \bar{u}^{\mathrm{cpu}}(t)\right)^2, \tag{24}$$

$$L^{\mathrm{mem}}(t) = \frac{1}{|\bar{\mathcal{N}}^s|} \sum_{n \in \bar{\mathcal{N}}^s} \left(u_n^{\mathrm{mem}}(t) - \bar{u}^{\mathrm{mem}}(t)\right)^2, \tag{25}$$

$$L^{\mathrm{bw}}(t) = \frac{1}{|\bar{\mathcal{A}}^s|} \sum_{(i,j) \in \bar{\mathcal{A}}^s} \left(u_{ij}^{\mathrm{bw}}(t) - \bar{u}^{\mathrm{bw}}(t)\right)^2. \tag{26}$$

Hence, the overall load-imbalance measure is defined as

$$L^{\mathrm{tot}}(t) = \omega_1 L^{\mathrm{cpu}}(t) + \omega_2 L^{\mathrm{mem}}(t) + \omega_3 L^{\mathrm{bw}}(t), \tag{27}$$

where $\omega_1, \omega_2, \omega_3 \geq 0$ and $\omega_1 + \omega_2 + \omega_3 = 1$.

7) Revenue and operational cost The revenue obtained at time slot t is defined as

$$\mathrm{Rev}(t) = \sum_{r \in \mathcal{R}_t} \eta_r \left(\rho_c \sum_{u \in \mathcal{N}_r^v} c_{r,u} + \rho_m \sum_{u \in \mathcal{N}_r^v} m_{r,u} + \rho_b \sum_{e \in \mathcal{L}_r^v} b_{r,e} \right), \tag{28}$$

where ρ_c, ρ_m, and ρ_b denote the unit revenues of CPU, memory, and bandwidth resources, respectively.

The total operational cost at time slot t is then given by

$$\mathrm{Cost}(t) = \sum_{n \in \bar{\mathcal{N}}^{\mathrm{dy}}} \left(E_n^{\mathrm{run}}(t) + E_n^{\mathrm{sw}}(t) \right) + E^{\mathrm{top}}(t) + C^{\mathrm{mig}}(t) + C^{\mathrm{res}}(t). \tag{29}$$

1.5 Optimization Objective

The considered dynamic embedding problem jointly determines request admission, virtual-node placement, virtual-link routing, node migration, dynamic-node operation, and dynamic-topology reconfiguration over the entire planning horizon. To this end, we maximize the long-term normalized utility

$$\max_{\mathbf{x},\mathbf{f},\mathbf{g},\mathbf{z},\mathbf{q},\boldsymbol{\eta}} \frac{1}{T} \sum_{t=1}^{T} \left[\lambda_1 \frac{\mathrm{Rev}(t)}{\mathrm{Rev}^{\max}} - \lambda_2 \frac{\mathrm{Cost}(t)}{\mathrm{Cost}^{\max}} - \lambda_3 \frac{L^{\mathrm{tot}}(t)}{L^{\max}} \right], \tag{30}$$

where $\lambda_1, \lambda_2, \lambda_3 \geq 0$ and $\lambda_1 + \lambda_2 + \lambda_3 = 1$. Here, $\mathrm{Rev}^{\max}$, $\mathrm{Cost}^{\max}$, and $L^{\max}$ are normalization constants.

The objective in (30) captures the tradeoff among three tightly coupled criteria: i) the long-term revenue obtained from admitted virtual network requests, ii) the overall operational cost, including node running energy, node switching energy, dynamic-topology energy, and migration cost, and iii) the system-wide load-balancing performance across node and link resources. Compared with conventional static-topology VNE formulations, the proposed model additionally determines when to power on or power off dynamic substrate nodes, when to migrate virtual nodes, and when to activate or deactivate dynamic links.

1.6 Constraints

The joint optimization is subject to the following constraints:

$$\sum_{n\in\bar{\mathcal{N}}^s} x_{r,u,n}(t) = \eta_r, \qquad \forall t\in\mathcal{T},\ \forall r\in\mathcal{R}_t,\ \forall u\in\mathcal{N}_r^v. \tag{31}$$

$$x_{r,u,n}(t) \le \tilde{z}_n(t), \qquad \forall t\in\mathcal{T},\ \forall r\in\mathcal{R}_t,\ \forall u\in\mathcal{N}_r^v,\ \forall n\in\bar{\mathcal{N}}^s, \tag{32}$$

where

$$\tilde{z}_n(t) = \begin{cases} 1, & n\in\mathcal{N}^{\mathrm{st}}, \\ z_n(t), & n\in\bar{\mathcal{N}}^{\mathrm{dy}}. \end{cases}$$

$$\sum_{r\in\mathcal{R}_t}\sum_{u\in\mathcal{N}_r^v} c_{r,u}\, x_{r,u,n}(t) \le \overline{C}_n\, \tilde{z}_n(t), \qquad \forall t\in\mathcal{T},\ \forall n\in\bar{\mathcal{N}}^s. \tag{33}$$

$$\sum_{r\in\mathcal{R}_t}\sum_{u\in\mathcal{N}_r^v} m_{r,u}\, x_{r,u,n}(t) \le \overline{M}_n\, \tilde{z}_n(t), \qquad \forall t\in\mathcal{T},\ \forall n\in\bar{\mathcal{N}}^s. \tag{34}$$

$$\sum_{j:(i,j)\in\bar{\mathcal{A}}^s} f_{r,e,ij}(t) - \sum_{j:(j,i)\in\bar{\mathcal{A}}^s} f_{r,e,ji}(t) = x_{r,u,i}(t) - x_{r,v,i}(t), \qquad \forall t\in\mathcal{T},\ \forall r\in\mathcal{R}_t,\ \forall e=(u,v)\in\mathcal{L}_r^v,\ \forall i\in\bar{\mathcal{N}}^s. \tag{35}$$

$$f_{r,e,ij}(t) \le y_{ij}(t), \qquad \forall t\in\mathcal{T},\ \forall r\in\mathcal{R}_t,\ \forall e\in\mathcal{L}_r^v,\ \forall (i,j)\in\bar{\mathcal{A}}^s. \tag{36}$$

$$\sum_{r\in\mathcal{R}_t}\sum_{e\in\mathcal{L}_r^v} b_{r,e}\, f_{r,e,ij}(t) \le \overline{B}_{ij}\, y_{ij}(t), \qquad \forall t\in\mathcal{T},\ \forall (i,j)\in\bar{\mathcal{A}}^s. \tag{37}$$

$$\sum_{\ell\in\bar{\mathcal{L}}^{\mathrm{dy}}:\ n\in\ell} q_\ell(t) \le \Pi_n, \qquad \forall t\in\mathcal{T},\ \forall n\in\bar{\mathcal{N}}^s. \tag{38}$$

$$\sum_{\ell\in\bar{\mathcal{L}}^{\mathrm{dy}}} q_\ell(t) \le Q^{\max}, \qquad \forall t\in\mathcal{T}. \tag{39}$$

$$\delta_n^+(t) + \delta_n^-(t) \le 1, \qquad \forall t=2,\dots,T,\ \forall n\in\bar{\mathcal{N}}^{\mathrm{dy}}. \tag{40}$$

$$z_n(t) - z_n(t-1) = \delta_n^+(t) - \delta_n^-(t), \qquad \forall t=2,\dots,T,\ \forall n\in\bar{\mathcal{N}}^{\mathrm{dy}}. \tag{41}$$

$$\sigma_\ell^+(t) + \sigma_\ell^-(t) \le 1, \qquad \forall t=2,\dots,T,\ \forall \ell\in\bar{\mathcal{L}}^{\mathrm{dy}}. \tag{42}$$

$$q_\ell(t) - q_\ell(t-1) = \sigma_\ell^+(t) - \sigma_\ell^-(t), \qquad \forall t=2,\dots,T,\ \forall \ell\in\bar{\mathcal{L}}^{\mathrm{dy}}. \tag{43}$$

$$g_{r,u,nm}(t) \ge x_{r,u,n}(t-1) + x_{r,u,m}(t) - 1, \qquad \forall t=2,\dots,T,\ \forall r\in\mathcal{R}_t,\ \forall u\in\mathcal{N}_r^v,\ \forall n,m\in\bar{\mathcal{N}}^s,\ n\neq m. \tag{44}$$

$$g_{r,u,nm}(t) \le x_{r,u,n}(t-1), \qquad \forall t=2,\dots,T,\ \forall r\in\mathcal{R}_t,\ \forall u\in\mathcal{N}_r^v,\ \forall n,m\in\bar{\mathcal{N}}^s,\ n\neq m. \tag{45}$$

$$g_{r,u,nm}(t) \le x_{r,u,m}(t), \qquad \forall t=2,\dots,T,\ \forall r\in\mathcal{R}_t,\ \forall u\in\mathcal{N}_r^v,\ \forall n,m\in\bar{\mathcal{N}}^s,\ n\neq m. \tag{46}$$

$$\sum_{\substack{n,m\in\bar{\mathcal{N}}^s \\ n\neq m}} g_{r,u,nm}(t) \le 1, \qquad \forall t=2,\dots,T,\ \forall r\in\mathcal{R}_t,\ \forall u\in\mathcal{N}_r^v. \tag{47}$$

$$\sum_{r\in\mathcal{R}_t}\sum_{u\in\mathcal{N}_r^v}\sum_{\substack{n,m\in\bar{\mathcal{N}}^s \\ n\neq m}} g_{r,u,nm}(t) \le \Gamma^{\max}, \qquad \forall t=2,\dots,T. \tag{48}$$

$$\sum_{n \in \bar{\mathcal{N}}^{\mathrm{dy}}} \delta_n^+(t) + \sum_{\ell \in \bar{\mathcal{L}}^{\mathrm{dy}}} \sigma_\ell^+(t) \leq \Omega_t, \qquad \forall t = 2, \ldots, T. \tag{49}$$

Constraints (31)–(34) govern virtual-node placement. Specifically, (31) ensures that each virtual node of an admitted request is assigned to exactly one candidate substrate node at each time slot, while (32) enforces node-state feasibility. Constraints (33) and (34) guarantee that the aggregate CPU and memory demands allocated to each substrate node do not exceed its available capacities. Constraints (35)–(37) describe virtual-link routing feasibility. In particular, (35) imposes flow conservation for each embedded virtual link, (36) ensures that routing can only use active substrate arcs, and (37) enforces arc bandwidth limits. Constraints (38) and (39) restrict the scale of the reconfigurable topology by limiting, respectively, the number of incident active dynamic links at each substrate node and the total number of active dynamic links in the system. Constraints (40) and (41) characterize dynamic-node switching decisions, while (42) and (43) characterize dynamic-link switching decisions. These constraints ensure consistent state transitions across consecutive time slots. Constraints (44)–(48) model virtual-node migration across time. Constraints (44)–(46) identify whether a migration occurs between two consecutive time slots, (47) ensures that each virtual node experiences at most one migration per time slot, and (48) limits the total migration intensity of the system. Finally, constraint (49) bounds the total amount of reconfiguration that can be performed in each time slot.